

Nominal and Real GDP

- ▶ **Nominal GDP** values current production at current prices.
- ▶ **Real GDP** adjusts for price changes. In our example, we value each year's quantities at base year prices.
- ▶ Higher spending can reflect more output, higher prices, or both.

If medicine sales rise, have we necessarily produced more medicine?

GDP per Person

$$\text{Real GDP per person} = \frac{\text{Real GDP}}{\text{Population}}$$

- ▶ Use GDP per person to compare average output, rather than the size of an economy.
- ▶ An average does not show how income is distributed.
- ▶ International comparisons also need to account for differences in purchasing power.

Would a larger economy necessarily have higher output per person?

Calculating a Price Index

For one medicine package of unchanged size and quality:

$$\text{Price index} = \frac{\text{Current year price}}{\text{Base year price}} \times 100$$

The package costs \$100 in the base year and \$120 in the current year.

$$\text{Current year index} = \frac{120}{100} \times 100 = 120$$

- ▶ The **base year index is 100**. An index of 120 means the price is **20% higher** than in the base year, not 120% higher.
- ▶ An index is not a dollar price. Next, we combine many goods and services in a consumer basket.

Classroom example. All dollar values are hypothetical.

The Consumer Price Index

The **CPI** measures changes in the prices of goods and services bought by consumers.

1. Choose a representative basket and its quantities.
2. Calculate the cost of buying that basket in each year.
3. Compare each year's cost with the base year cost.

$$\text{CPI} = \frac{\text{Current cost of the basket}}{\text{Base year cost of the basket}} \times 100$$

The GDP Deflator

The **GDP deflator** measures prices of domestically produced final output.

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

Using our medicine and laboratory test economy:

$$\frac{30,100}{26,000} \times 100 \approx 115.77$$

- ▶ The CPI was 115. It held the Year 1 consumer basket fixed.
- ▶ This deflator uses the Year 2 output mix. Different weights give a different result.

Classroom example. All dollar values are hypothetical.

Comparing Price Measures

Measure	Prices measured	Medicine example
CPI	Consumer goods and services, including imports	Retail purchase by a household
PPI	Selling prices received by domestic producers	Sale by a medicine producer
GDP deflator	Domestically produced final output, including exports	Medicine produced for export

A producer's selling price and a household's retail price can change differently.

Source: BLS, CPI and PPI overviews; BEA, GDP price measures.

Calculating a Price Index

One medicine package of unchanged size and quality costs \$100 in the base year and \$120 now.

$$\text{Price index} = \frac{\text{Current price}}{\text{Base year price}} \times 100 = \frac{120}{100} \times 100 = 120$$

- ▶ The base year index is **100**. An index of 120 means the price is **20% higher**, not 120% higher.
- ▶ An index has no currency unit. To measure consumer prices more broadly, combine purchases in a basket.

The Consumer Price Index

The **CPI** measures changes in the prices of a representative basket of consumer goods and services.

$$\text{CPI} = \frac{\text{Current cost of the fixed basket}}{\text{Base year cost of the fixed basket}} \times 100$$

- ▶ Holding quantities and quality fixed isolates the effect of changing prices.
- ▶ Housing includes rent and the rental value of owner-occupied housing services; buying a house is an asset purchase.
- ▶ Brazil's **IPCA** is also a consumer price index, with different coverage and methods from the U.S. CPI.

Source: BLS, CPI concepts; IBGE, inflation.

Inflation and the Price Level

$$\text{Inflation rate} = \frac{\text{New CPI} - \text{Previous CPI}}{\text{Previous CPI}} \times 100$$

	Year 1	Year 2	Year 3
CPI	100	115	120.75
Inflation from previous year		15%	5%

Inflation slows from 15% to 5%. Have prices fallen?

No. The basket still costs more than in Year 2.

Disinflation: slower price growth. **Deflation:** falling prices.

The GDP Deflator

The **GDP deflator** measures prices of domestically produced final output.

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100 = \frac{30,100}{26,000} \times 100 \approx 115.77$$

- ▶ Our medicine and laboratory test economy values the **same current output** at current prices for nominal GDP and base prices for real GDP.
- ▶ The CPI held the Year 1 consumer basket fixed. This deflator uses the Year 2 output mix, with more weight on medicine.
- ▶ Different coverage and weights can produce different price indexes.

Source: BEA, GDP deflator.

Reading an Inflation Report

- ▶ **Monthly inflation** compares a month with the previous month. **Twelve month inflation** compares it with the same month one year earlier.
- ▶ **Headline CPI** includes all items. **Core CPI** excludes food and energy to help identify movements beyond those volatile categories.
- ▶ Core inflation does not describe a household budget without food or energy. Those purchases still matter for living costs.

Source: BLS, CPI FAQ.